

PORTFOLIO PROJECT • E-COMMERCE ANALYSIS

Beyond the Dashboard

RFM Segmentation Tool With Excel

Champions

Loyal

Potential

At Risk

The Source Dataset

Every model, chart, and insight in this deck (RFM segmentation, product, price, geography, and trend analysis) is built entirely from one raw transaction data file. A total of 531,147 Rows x 8 Cols. were analyzed.

- TransactionNo & Date — order-level identifiers
- ProductNo & ProductName — item sold
- Price & Quantity — line-level revenue inputs
- CustomerNo & Country — who bought, and from where

Original dataset on Kaggle

An Online Shop Business — Gabriel Ramos

kaggle.com/datasets/gabrielramos87/an-online-shop-business

	A	B	C	D	E	F	G	H	I
1	TransactionNo	Date	ProductNo	ProductName	Price	Quantity	CustomerNo	Country	
2	581482	12/9/2019	22485	Set Of 2 Wooden Market Crates	21.47	12	17490	United Kingdom	
3	581475	12/9/2019	22596	Christmas Star Wish List Chalkboard	10.65	36	13069	United Kingdom	
4	581475	12/9/2019	23235	Storage Tin Vintage Leaf	11.53	12	13069	United Kingdom	
5	581475	12/9/2019	23272	Tree T-Light Holder Willie Winkie	10.65	12	13069	United Kingdom	
6	581475	12/9/2019	23239	Set Of 4 Knick Knack Tins Poppies	11.94	6	13069	United Kingdom	
7	581475	12/9/2019	21705	Bag 500g Swirly Marbles	10.65	24	13069	United Kingdom	
8	581475	12/9/2019	22118	Joy Wooden Block Letters	11.53	18	13069	United Kingdom	
9	581475	12/9/2019	22119	Peace Wooden Block Letters	12.25	12	13069	United Kingdom	
10	581475	12/9/2019	22217	T-Light Holder Hanging Lace	10.65	12	13069	United Kingdom	
11	581475	12/9/2019	22216	T-Light Holder White Lace	10.55	24	13069	United Kingdom	
12	581475	12/9/2019	22380	Toy Tidy Spaceboy	11.06	20	13069	United Kingdom	
13	581475	12/9/2019	22442	Grow Your Own Flowers Set Of 3	12.25	12	13069	United Kingdom	
14	581475	12/9/2019	22664	Toy Tidy Dolly Girl Design	11.06	20	13069	United Kingdom	
15	581475	12/9/2019	22721	Set Of 3 Cake Tins Sketchbook	12.25	12	13069	United Kingdom	
16	581475	12/9/2019	22723	Set Of 6 Herb Tins Sketchbook	11.53	12	13069	United Kingdom	
17	581475	12/9/2019	22785	Squarecushion Cover Pink Union Jack	11.53	12	13069	United Kingdom	
18	581475	12/9/2019	22955	36 Foil Star Cake Cases	11.06	24	13069	United Kingdom	
19	581475	12/9/2019	23141	Triple Wire Hook Pink Heart	11.06	12	13069	United Kingdom	
20	581475	12/9/2019	22956	36 Foil Heart Cake Cases	11.06	24	13069	United Kingdom	
21	581475	12/9/2019	22581	Wood Stocking Christmas Scandisport	10.55	48	13069	United Kingdom	
22	581476	12/9/2019	23198	Pantry Magnetic Shopping List	11.53	48	12433	Norway	
23	581476	12/9/2019	23197	Sketchbook Magnetic Shopping List	11.74	24	12433	Norway	
24	581476	12/9/2019	23184	Bull Dog Bottle Opener	15.32	8	12433	Norway	
25	581476	12/9/2019	23168	Classic Cafe Sugar Dispenser	11.53	12	12433	Norway	
26	581476	12/9/2019	23167	Small Ceramic Tin Storage Jar	10.96	96	12433	Norway	

Raw transaction log — Sales Transaction sheet (v.4a)

Download Excel Project File: [GitHub Link](#)

Executive Summary



£62.3M

Total revenue generated across the full transaction history.



4,738

Unique customers, with a 98% repeat-purchase rate.



95% / 85%

Share of customers / revenue coming from the UK alone.



389 → 42%

Champions are 8% of customers but drive 42% of revenue.



£699K

Revenue sitting with 34 high-value **At-Risk customers.**



35%

Share of the year's revenue happens in Q4 alone.

Meet the Segmentation Tool

Every customer is scored 0–10 on three axes: how recently they bought, how often, and how much they spent, using percentile ranks. The shown pivot runs on a data model and using the slicer you can find customers by their respective segments.

Champions

High Recency + High Frequency + High Monetary — recent, frequent, high-value.

Loyal

High Frequency + High Monetary — steady, moderate-recency value.

Potential

High Recency, lower Frequency/Monetary — new buyers who haven’t spent much

At Risk

Low Recency + High Frequency + High Monetary — big spenders gone quiet.

Download Excel Project File: [GitHub Link](#)

Customer ID	Total Quantity Purchased	Total Transactions	Avg Order Value (AOV)	Total Revenue Generated
13093	3976	169	£3,371	£43,829
13162	2712	170	£5,340	£32,038
12378	2529	219	£29,411	£29,411
13958	2323	512	£14,185	£28,371
15379	2317	196	£3,278	£26,222
15823	1771	611	£8,122	£24,365
13869	1995	399	£4,840	£24,202
17850	1746	305	£668	£23,390
12071	1638	647	£23,190	£23,190
12237	1607	596	£22,960	£22,960
17926	1880	461	£5,602	£22,407
16546	1800	170	£5,333	£21,330
16349	1511	627	£10,628	£21,255
15638	1495	599	£10,433	£20,867
12026	1679	409	£20,486	£20,486
15104	1579	427	£4,859	£19,435
17177	1310	537	£9,439	£18,877
15424	1501	371	£6,073	£18,218
14977	1387	416	£17,353	£17,353
17544	1205	524	£17,340	£17,340
16055	1393	494	£8,637	£17,274
16919	1403	326	£1,547	£17,013
14614	1406	384	£16,524	£16,524
17646	1300	402	£3,203	£16,466

How to Interpret the Results? - Focus on Customer Segment + Interpretation Columns			
RFM Basics		Score Interpretation	
RFM Component	What it measures		
Recency (R)	How recently the customer purchased	10 = most recent, 1 = least recent	
Frequency (F)	How often the customer purchased	10 = most frequent, 1 = least frequent	
Monetary (M)	How much the customer spent	10 = highest spending, 1 = lowest spending	
Customer Segmentation Logic			
Customer Segment	Logic	Interpretation	
Champions	High R + High F + High M	Recent, frequent and high-value customers	
Loyal	High R + Moderate F + High M	High-value customers with moderate purchase frequency	
Potential	High R + Lower F + Lower M	Recent customers who are starting to buy	
At Risk	Low R + High F + High M	High-value, frequent customers who stopped purchasing recently	

customer_segmentation

At Risk

Champions

Not segmented

Potential Customers

Straight from the workbook: the live segmentation tool screenshot.



Use Case: Finding Revenue at Risk

Using the slicer, you will see a list of customers at risk. These are customers that **used to spend a lot but have now gone quiet**. The results are sorted by the total revenue column and top ten values are highlighted. This data can be used by marketing and retention teams.

34

at-risk customers identified

£699K

combined revenue exposure

£20.6K

avg. value per customer — 57% above the typical customer

~230 days

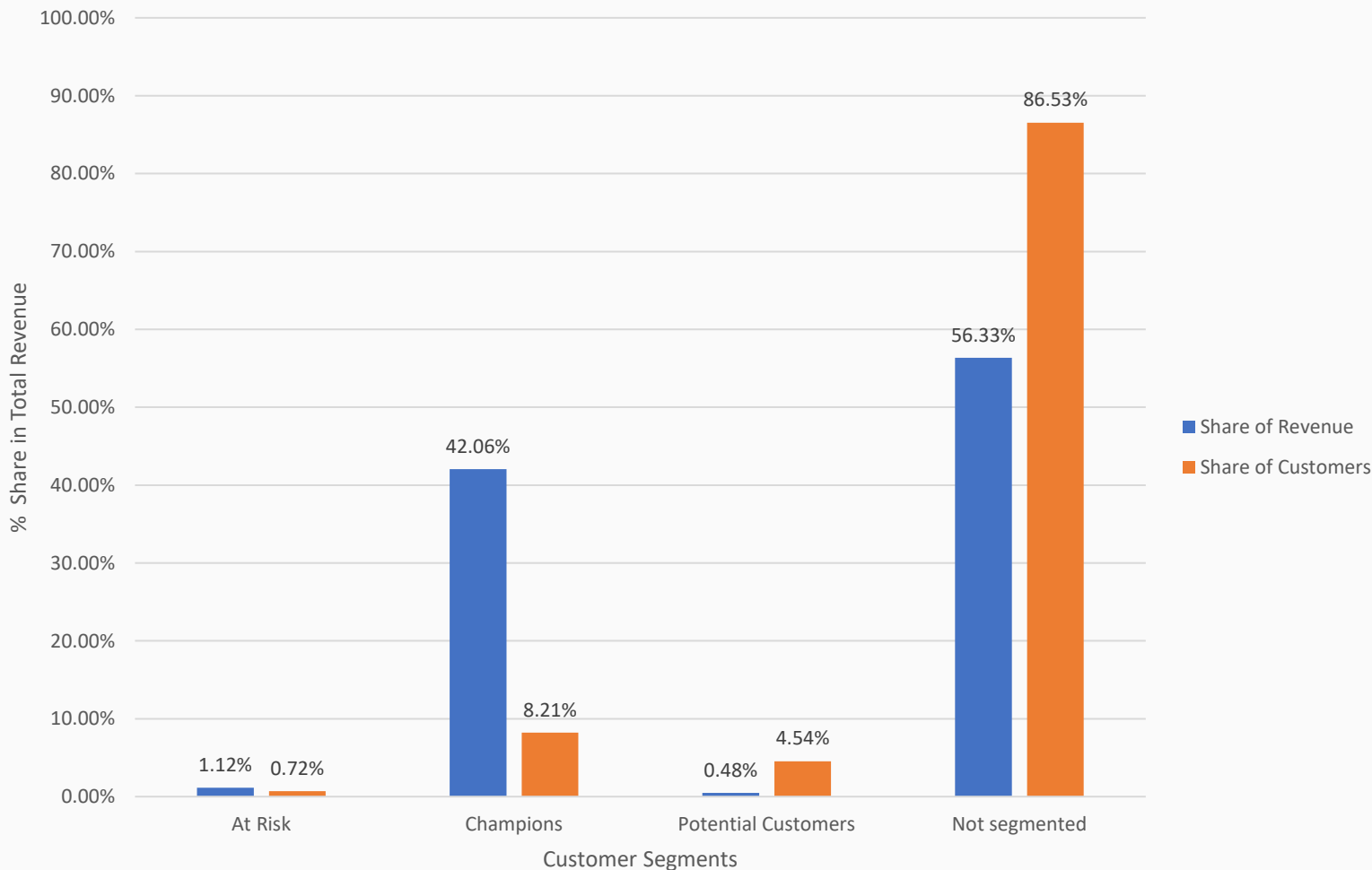
average time since their last order

Customer ID	Total Quantity Purchased	Total Transactions	Avg Order Value (AOV)	Total Revenue Generated
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16919	1403	326	£1,547	£17,013
14614	1406	384	£16,524	£16,524
17646	1309	402	£3,293	£16,466
14453	1168	478	£8,231	£16,461
15808	1307	197	£3,255	£16,276

These are historically high-frequency, high-spend customers who've gone quiet: a ready-made target list for a win-back campaign.

Small Segments, Outsized Impact

Share of Customers vs. Share of Revenue by Segment



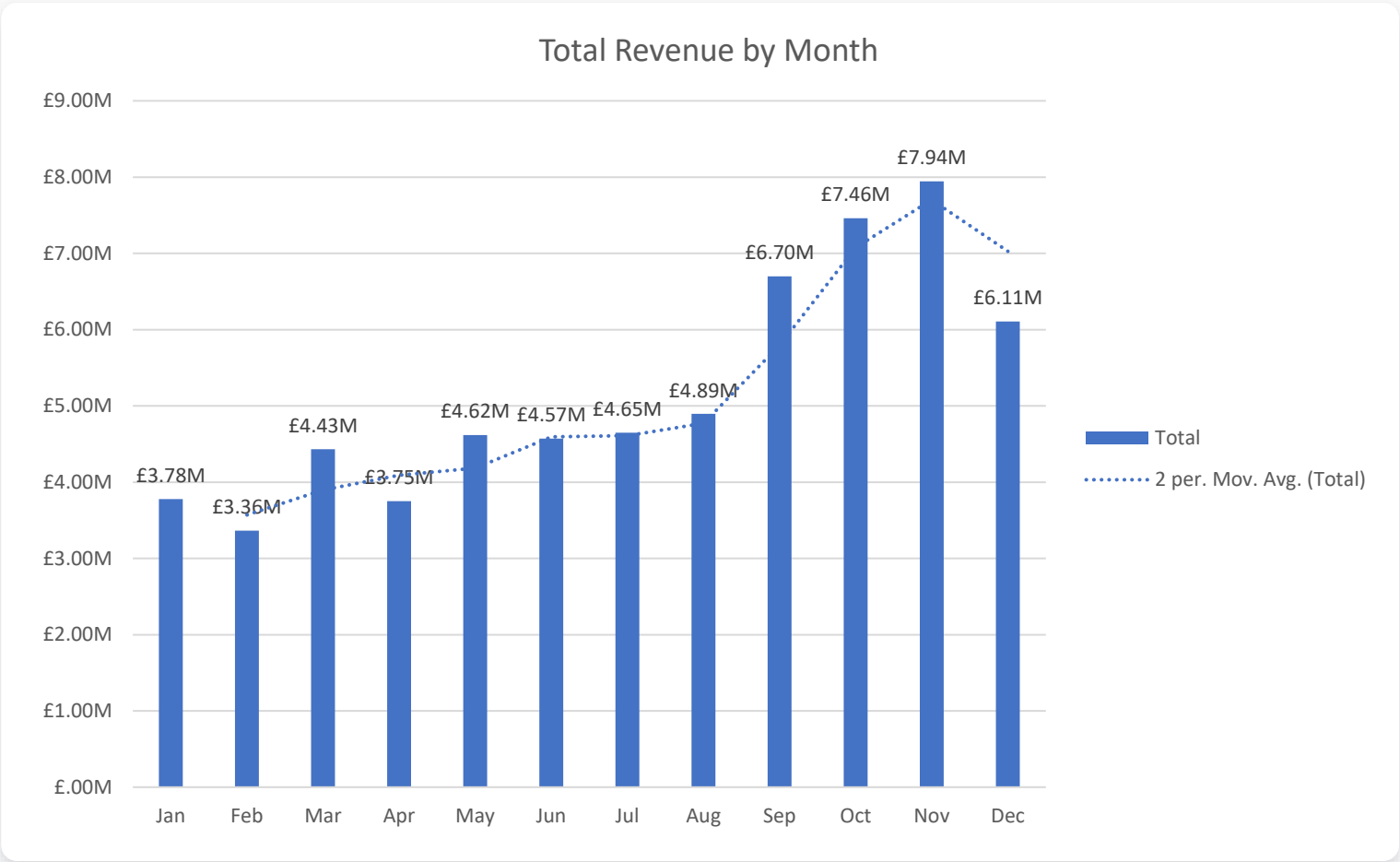
Why it matters

Champions are just 8% of customers but 42% of revenue: value them first.

Not Segmented are left out customers which can be segmented further using RFM, as per need.

Potential & At Risk are small in number, but the clearest growth and retention levers on the board.

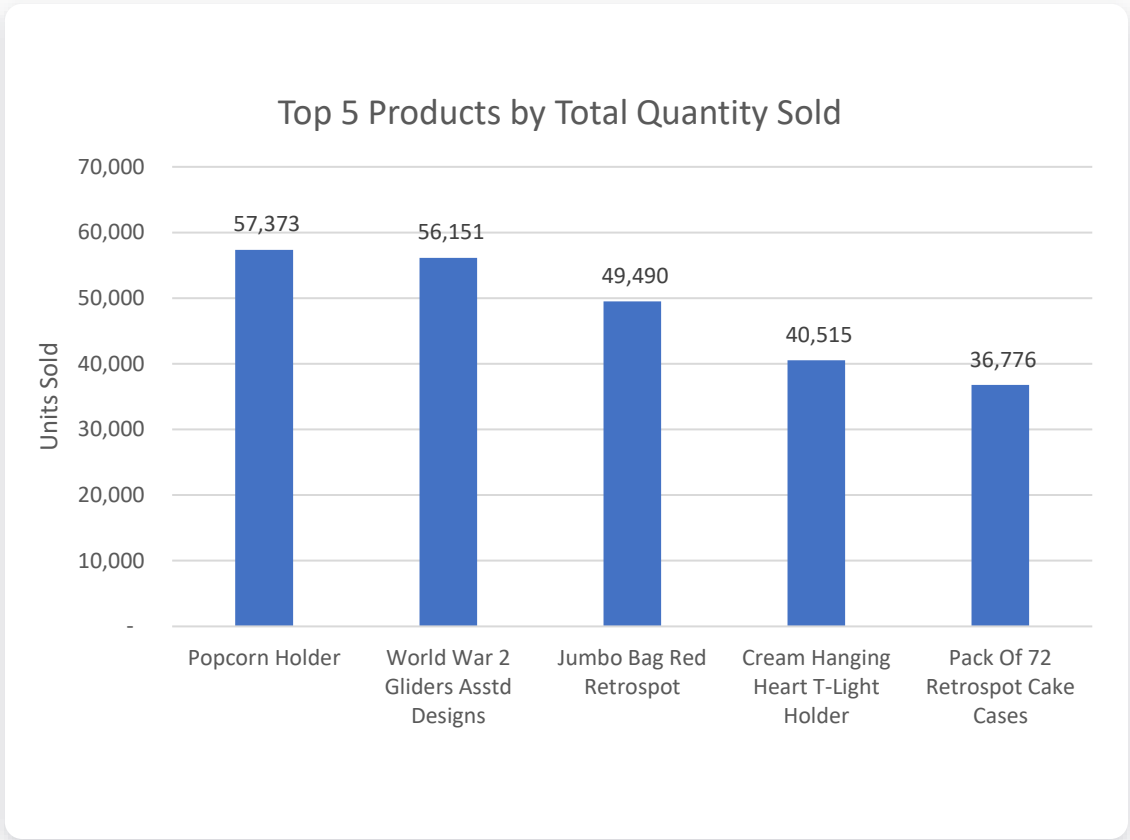
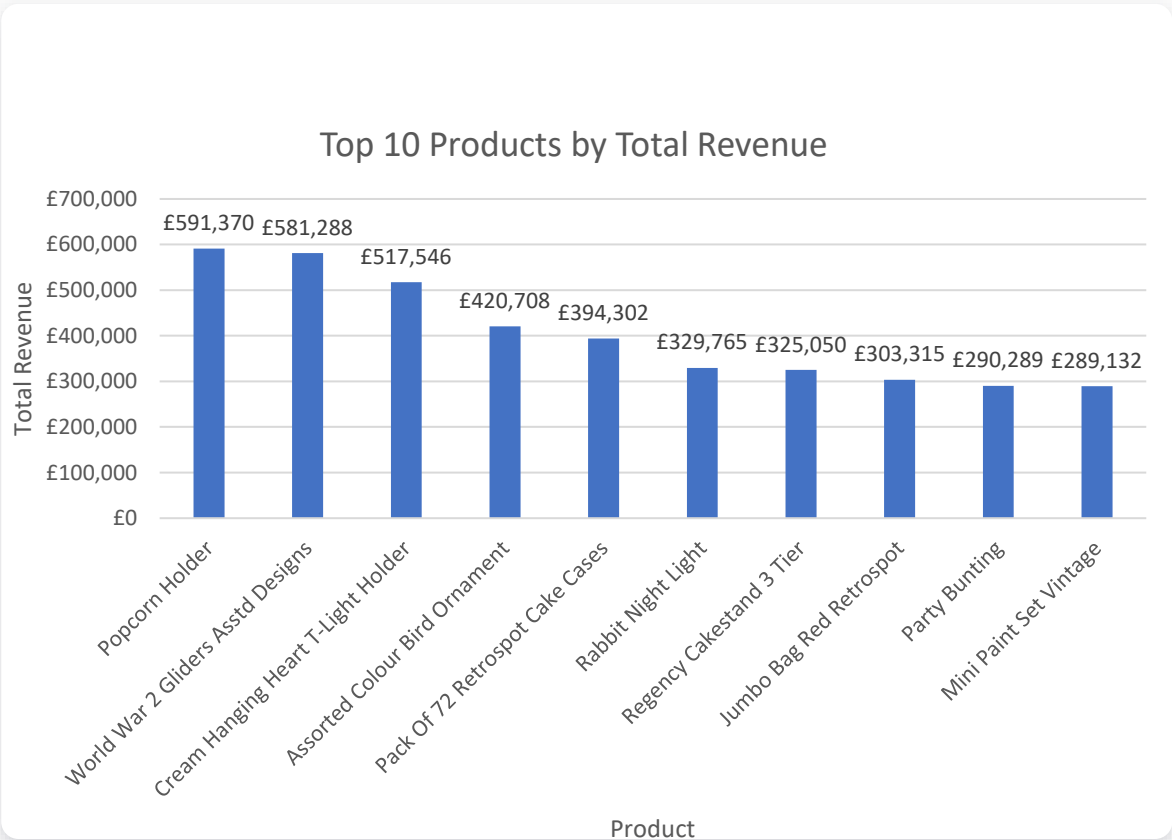
The Quarter 4 Surge



Month	Total Revenue
Jan	↓ £3.78M
Feb	↓ £3.36M
Mar	↓ £4.43M
Apr	↓ £3.75M
May	↓ £4.62M
Jun	↓ £4.57M
Jul	↓ £4.65M
Aug	→ £4.89M
Sep	↑ £6.70M
Oct	↑ £7.46M
Nov	↑ £7.94M
Dec	→ £6.11M
Grand Total	£62.26M

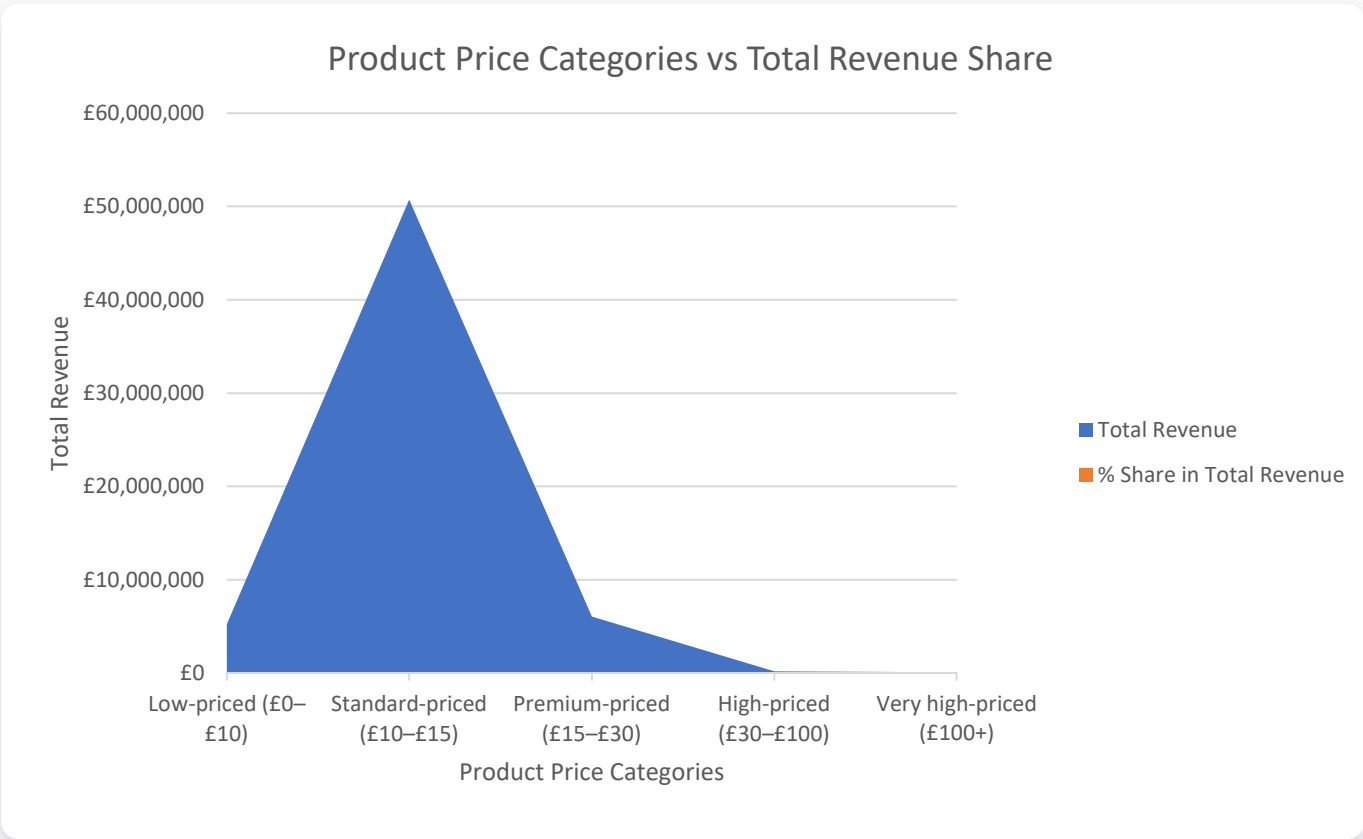
Q4 alone brings in £21.5M (35% of annual revenue) and 8,262 transactions (36% of annual volume): November is the single biggest month at £7.94M. This can help prepare for future surges in Q4.

What's Actually Selling



Popcorn Holder tops both charts: highest revenue (£591K) and highest volume (57K units). But the top 10 products combined are only 6.5% of total revenue: this is a long-tail catalog, not a hits-driven one.

The £10–£15 Sweet Spot



81%

of total revenue (£50.7M) comes from the Standard-priced band — £10 to £15. Median product price sits at £11.94.

Price Ranges	Total Revenue	% Share in Total Revenue
Low-priced (£0–£10)	£5,216,539	8.38%
Standard-priced (£10–£15)	£50,741,583	81.50%
Premium-priced (£15–£30)	£6,031,596	9.69%
High-priced (£30–£100)	£193,810	0.31%
Very high-priced (£100+)	£78,182	0.13%

Pricing outside the £10–£15 band barely moves revenue: a strong signal for where future SKUs should be priced.

A UK Business, Mostly

Country	Count of Custom	Percentag of Custome
United Kingdom	4324	95.07%
Germany	92	2.02%
France	87	1.91%
Spain	23	0.51%
Belgium	22	0.48%

95%

of customers and 85% of revenue (£51.7M) comes from the United Kingdom.

Country	Total Revenue	Total Revenue	Avg Revenue Per Cu	Count of Customers
United Kingdom	£51,671,892	85.13%	10906	4324
Netherlands	£2,155,296	3.55%	455	9
Ireland	£1,764,585	2.91%	372	13
Germany	£1,391,117	2.29%	294	92
France	£1,343,662	2.21%	284	87
Australia	£1,002,266	1.65%	212	9
Sweden	£407,717	0.67%	86	7
Switzerland	£365,238	0.60%	77	21
Japan	£303,017	0.50%	64	8
Spain	£296,117	0.49%	62	23

Netherlands has the highest avg. revenue per customer outside the UK (£455): a small but efficient market worth a closer look.

A Remarkably Sticky Customer Base

98%

REPEAT-PURCHASE RATE

Only 102 of 4,738 customers (2%) are one-time buyers. Once someone converts, they almost always come back and retention is clearly this business's strength.

Customer Type ▾	Customer Count	Count of customer_no	Total Transactions
One-time customer	102	2%	102
Repeat-customer	4636	98%	531,044
Grand Total	4738	100.00%	531146

The flip side: with so few first-time buyers, top-of-funnel acquisition looks thin which raises a natural next question for this dataset.

Behind the Scenes: RFM Logic (IFS)

1. Score Recency, Frequency and Monetary are each converted to a 0–10 score using PERCENTRANK.EXC against the full customer base.

2. Classify A nested IFS() formula reads the three scores in order and assigns the first matching segment which are Champions, Loyal, Potential, or At Risk.

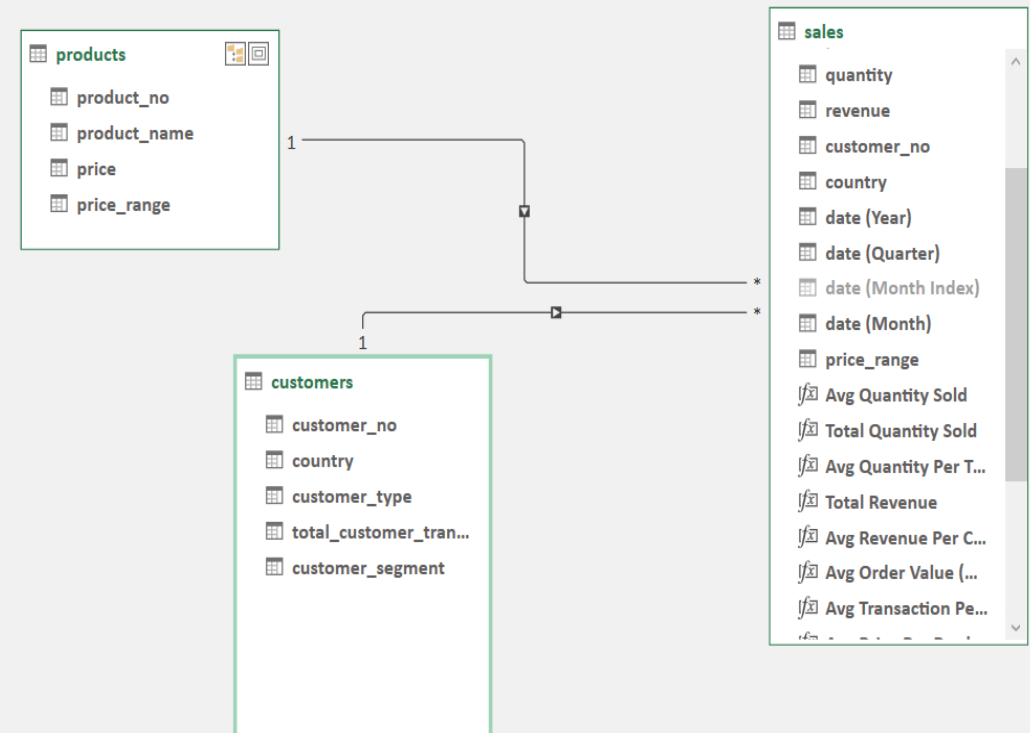
3. Fall through Anyone who doesn't clear a segment's thresholds is labeled Not Segmented as the long tail seen on slide 6.

```
=IFS(AND('rfm_logic'!$F2>= 8, 'rfm_logic'!$G2 >= 8, 'rfm_logic'!$H2 >= 8), "Champions", AND(AND('rfm_logic'!$F2 >=7, 'rfm_logic'!$F2 <= 5), 'rfm_logic'!$G2 >=8, 'rfm_logic'!$H2 >= 8), "Loyal Customers", AND('rfm_logic'!$F2 >= 8, 'rfm_logic'!$G2 <= 3, 'rfm_logic'!$H2 <= 3), "Potential Customers", AND('rfm_logic'!$F2<=3, 'rfm_logic'!$G2 >= 8, 'rfm_logic'!$H2 >= 8), "At Risk", TRUE, "Not segmented")
```

customer_id	most_recent_order_date	days_since_last_order	count_of_orders	total_revenue	recency	frequency	monetary	customer_segment
14646	12/8/2019	1	2066	2115604	10	9	9	9 Champions
14911	12/8/2019	1	5794	955800	10	9	9	9 Champions
12415	11/15/2019	24	776	905824	7	9	9	9 Not segmented
17450	12/1/2019	8	339	905692	9	9	9	9 Champions
18102	12/9/2019	0	431	897137	10	9	9	9 Champions
14156	11/30/2019	9	1399	705298	9	9	9	9 Champions
13694	12/6/2019	3	803	660262	10	9	9	9 Champions
17511	12/7/2019	2	1074	650327	10	9	9	9 Champions
14298	12/6/2019	3	1638	637520	10	9	9	9 Champions
16684	12/5/2019	4	278	538198	10	9	9	9 Champions
16029	11/1/2019	38	260	508571	6	8	8	9 Not segmented
15311	12/9/2019	0	2451	397299	10	9	9	9 Champions
16422	11/22/2019	17	378	382656	8	9	9	9 Champions
17949	12/8/2019	1	76	382459	10	6	6	9 Not segmented

Behind the Scenes: Data Model

Raw sales, customers and products tables are linked by customer and product IDs, then rolled up into the analysis sheets behind every slide in this deck that calculates segmentation, product, price, country and trend. Using Power Pivot, the model and customized measures like AOV, Average Quantity Sold, etc. are also created.



Thank You

Questions? Let's talk about what the data found and what to do next.

Champions

Loyal

Potential

At Risk